

Database Management Systems – MS Access

A Database is a collection of information related to a particular subject or purpose.

Issues in the use of a Database Management System:

- ☐ Efficiency
- ☐ Robustness (Strong)
- ☐ Security
- ☐ Capacity
- ☐ Cost
- ☐ Accuracy

DBMS Software Packages:

- ☐ Dbase III+
- ☐ FoxPro
- ☐ Microsoft Access

Microsoft Access

Creating a Database:

Open Access → Click on **Blank Desktop Database** → Type a **File Name** for the database → Select the location to save the database by clicking the **Browse** button → **Create**

Using MS Access, you can manage all your information from a single database file. Within this file you divide your data into separate storage containers. An Access database is made of six containers.

1. Tables
2. Queries
3. Forms
4. Reports
5. Macros
6. Modules

Tables:

A **Table** is a collection of data about a specific topic. It is made of **Fields** and **Records**.

Fields are separate columns of information that makeup a record within a table. Each **Field** should have a **Field Name**.

A **Record** is a description of a person or something in a table.

Field

1st Batch - 2017						
ID	AdNo	Group	Name	School	Address	TPNO
1	7651	A	H.A.Minoli Anjalika Hettiarachchi	Vihara Maha Devi B.M.V	405/33, Udupila North, Delgoda.	0777837720
2	7652	A	D.G. Shalshana Giyumi Thisara	Sapugaskanda M.V.	451/8, Makola North, Makola.	0711784370
3	7653	A	Heshani Anuththara Liyanage	Sapugaskanda M.V.	405/13, Udupila, Delgoda	0112978120
4	7654	A	D. Chethani Isini Dissanayake	Sapugaskanda M.V.	401/2/A, Makola North, Makola.	
5	7655	A	J.A. Pasan Dinakara	Sri Dheerananda Pirivena	348/2, Bandaranayake Mw; Heiyanthuduwa	0112487392
6	7656	A	P. Chathuni Prabodha Perera	Sangamitta M.V., Heiyanthuduwa	687/1, Kottunna Rd; Heiyanthuduwa	0714049269
7	7657	A	D. Sachini Kamasha Narangoda	Sangamitta M.V., Heiyanthuduwa	606/B-2, Nayagala, Heiyanthuduwa	0725812975
8	7658	A	L. Dilan Madushanka	Sapugaskanda M.V.	443/2, Makola North, Makola	0115675598
9	7659	A	W. Sethmini Samudika	Sapugaskanda M.V.	3/2, Siyambalape South, Siyambalape	0113147319

Record →

Creating a Table:

1. **Create** tab → **Tables** group → **Table** → Ignore **ID** Column → Click on the drop-down button on the **Click to Add** column → Select the Data type you want for the column → Rename the field name with the name you want → Likewise select data types and give field names as you want for the next columns also → **File** → **Save** → Type a name for the table → Enter data into the table
2. **Create** tab → **Tables** group → **Table Design** → Type the first **Field Name** → Go to the **Data Type** column → Click on the drop-down button and select the data type for your field → If you want type any **Description** → Likewise type other field names also → **Design** tab → **Views** group → **View** button → Click on the drop-down menu → **Datasheet View** → Click **Yes** to save the table → Type a table name → **OK** → Click **No** for the **Primary Key** → Enter data into the table

Change Views:

Home tab → **Views** group → **View** → Select any option

Data Types:

Type of Data	Description
Short Text (formerly known as “Text”)	Text or combinations of text and numbers, including numbers that do not require calculating (e.g. phone numbers).
Long Text (formerly known as “Memo”)	Lengthy text or combinations of text and numbers.
Number	Numeric data used in mathematical calculations
Date/Time	Date and time values for the years 100 through 9999.
Currency	Currency values and numeric data used in mathematical calculations involving data with one to four decimal places.
AutoNumber	A unique sequential (incremented by 1) number or random number assigned by Microsoft Access whenever a new record is added to a table.
Yes/No	Yes and No values and fields that contain only one of two values (Yes/No, True/False, or On/Off).

Attachment	Files, such as digital photos. Multiple files can be attached per record. This data type is not available in earlier versions of Access.
OLE objects	OLE objects can store pictures, audio, video, or other BLOBs (Binary Large Objects)
Hyperlink	Text or combinations of text and numbers stored as text and used as a hyperlink address.
Lookup Wizard	The Lookup Wizard entry in the Data Type column in the Design view is not actually a data type. When you choose this entry, a wizard starts to help you define either a simple or complex lookup field. A simple lookup field uses the contents of another table or a value list to validate the contents of a single value per row. A complex lookup field allows you to store multiple values of the same data type in each row.
Calculated	You can create an expression that uses data from one or more fields. You can designate different result data types from the expression.

Setting Field Properties:

Every table in Access is made up of fields. The properties of a field describe the characteristics and behavior of data added to that field.

1. **Field Size**

Determines the amount of space that is allocated for each value. For AutoNumber fields, only two values are allowed:

2. **Format**

You can use the **Format** property to customize the way numbers, dates, times, and text are displayed and printed.

3. **Decimal Places**

According to the format you have selected, you can get the desired number of **Decimal Places**

4. **Input Mask**

You can create input masks in Access to dictate a pattern used for data entry in

selected fields.

- 0 - A single digit must be entered in this position
- 9 - A digit or a space can be entered in this position
- # - A digit, a space, or a plus or minus sign can be entered in this position
- L - A letter must be entered in this position
- ? - A letter can be entered in this position
- A - A letter or a digit must be entered in this position
- & - A character or a space must be entered in this position
- C - Any character or a space can be entered in this position

5. Caption

If you enter a caption for the field, this caption will be displayed instead of the field name in all views apart from **Design view**.

6. Default Value

You add a default value to a table field or form control whenever you want Access to enter a value in a new record automatically.

7. Validation Rule

Validation rule is a field property that sets limits or conditions on the data that can be entered in the field.

Text validation rule examples

- Like "#####"
 - Like "????"
 - Like "K*"
 - Is Not Null
 - Not Null And Like "K*i*e"
 - Like "James Joyce"
 - Like "a?b?c"
 - Like "bi[nd]"
 - Not Like "[!a-z]*"
 - Not Like "[!a-z]*"
 - Not Like "[!0-9a-z]*"
 - "Jack" Or "John" Or "Marc"
- A number consisting of 8 digits
 - 4 characters
 - A text starting with the letter K or k
 - Any value but not null/empty
 - A text starting with the letter K/k, followed by zero or more characters (*), followed by an i/I, followed by zero or more characters (*), followed by an e. Null/empty is not allowed
 - Matches "James Joyce"
 - Matches an a, followed by any single character, followed by a b, followed by any single character, followed by a c
 - Matches "bin" or "bid"
 - Letters only
 - Letters and spaces only
 - Letters and numbers only
 - Jack, John or Marc

Number validation rule examples

Example	Matches...
> 0	Positive numbers only, excluding 0
>= 0	Positive numbers only, including 0
>= 0 And <= 100	Between 0 and 100, including 0 and 100
Between 0 And 100	Between 0 and 100, including 0 and 100
In(1,2,3,4,5)	1,2,3,4 or 5
1 Or 2 Or 3 Or 4 Or 5	1,2,3,4 or 5

8. Validation Text

Validation text is a custom message that's displayed if invalid data is entered in a **field** or control where a **validation rule** exists.

Opening a Database:

1. Open **MS Access** → Select the database if it is in the **Recent** list or if not open **Other Files** → **Browse** for the database you want
2. While **MS Access** is open, click on **File** tab → **Open** → **If the name of your database is displayed, click on it or Browse for the database**

Opening a Table:

Open the database → **All Access Objects** pane on the left side of the window → **Tables** → Double click on the table you want

A Primary Key:

A **primary key** is a special relational database table column (or combination of columns) designated to uniquely identify all table records. A **primary key's** main features are:

- It must contain a unique value for each row of data.
- It cannot contain null values.

Defining a Primary Key:

In order to connect information stored in separate tables, they must be included in a field that uniquely identified each individual record in the table. Such a field is called the “**Primary Key Field**”. You can use a primary key to identify each record in your table. MS Access has a special ability to bring together information stored in separate tables using queries, forms and reports. In order to do that, each table should include a

field that uniquely identifies each record stored in the table. MS Access prevents any duplicate or null values from being entered in the primary key field.

There are three kinds of primary keys that can be identified.

1. Auto Number Primary Key:

When you are going to save a table Access will ask you to create a primary key. If you answer **Yes**, Access will create an **Auto Number Primary Key**. The field name of that field will be “**ID**”.

2. Single Field Primary Key:

If you have a field that contains unique values such as **AdNo**, you can designate that field as a primary key.

3. Multiple Field Primary Key:

In situations like where you cannot guarantee the uniqueness of any single field, you can designate two or more fields as primary key fields.

Define a Primary Key:

1. Open the **Design View** of the table → Select the field/fields you want to define as the primary key → Right click on the selected field/fields → **Primary Key**
2. Open the **Design View** of the table → Select the field/fields you want to define as the primary key → **Design** tab → **Tools** group → **Primary Key**

Creating table relationships

A Relationship is how you tell the program that a piece of information means the same thing in more than one table. Table relationships can be created between two tables as long as they have a common field. Setting the relationship up in Access is very important. Once a relationship is established you can view information from both tables at the same time by creating a query.

Relationships are created using a **Primary Key** from one table and linking it to a same field in another table (now called a **Foreign Key**).

Types of table relationships

There are three types of table relationships.

□ **One-to-one relationship:**

Both of the common fields (typically the primary key and foreign key fields) must have a unique index. This means that the **Indexed** property for these fields should be set to **Yes (No Duplicates)**. If both fields have a unique index, Access creates a one-to-one relationship.

In a one-to-one relationship, each row in one database table is linked to 1 and only 1 other row in another table.

For example, in a library, one student can borrow only one book.

Student				Book Borrowing			
AdNo	Name	DOB	Grade	AdNo	BookNo	Book Name	Date issued
1110	Palitha Wimal	10/11/96	8	1110	801	Madolduwa	10/02/13
2910	Sriya Menike	02/20/97	10	2910	1002	Charitha Hathak	02/10/13
2220	Lalitha Perera	12/03/96	6	2220	906	English Grammar	14/04/13
1203	B.Mohommad	05/12/97	7	1203	712	Computer Studies	05/12/13
1230	T.Ramasinghe	21/11/97	6	1230	693	Electronics	20/10/13

AdNo in **Student** table is a **primary** key. **AdNo** in **Book Borrowing** table is a **Foreign** key. **AdNo** in both tables are indexed as: **Yes(No duplicates)**.

□ **One-to-many relationship:**

The field on one side (typically the primary key) of the relationship must have a unique index. This means that the **Indexed** property for this field should be set to **Yes (No Duplicates)**. The field on the many side should *not* have a unique index. It can have an index, but it must allow duplicates. This means that the **Indexed** property for this field should be set to either **No** or **Yes (Duplicates OK)**. When one field has a unique index, and the other does not, Access creates a one-to-many relationship.

In a one-to-many relationship, each row in the related to table can be related to many rows in the relating table.

For Example, in a library, one student can borrow maximum of three books.

Student				Book Borrowing			
Ad No	Name	DOB	Grade				
1110	Palitha Wimal	15/11/96	8				
2910	Sriya Menike	20/10/97	10				
2220	Lalitha Perera	12/03/96	6				
1203	B.Mohammed	05/12/97	7				
1230	T.Ramasinghe	21/11/97	6				

AdNo	BookNo	Book Name	Date issued
1110	801	Madolduwa	10/02/13
1110	1002	Charitha hathak	02/10/13
1110	906	English Grammar	14/04/13
1230	712	Computer Studies	05/12/13
1230	324	Computer H/W	22/04/13
1230	693	Electronics	20/10/13

AdNo in **Student** table is the primary key. **AdNo** in **Book Borrowing** table is the **Foreign key**. **AdNo** in Student table is indexed as: Yes (No duplicates) and **AdNo** in **Book Borrowing** table is: No or Yes (Duplicate OK)

Many-to-many relationship

In a many-to-many relationship, one or more rows in a table can be related to 0, 1 or many rows in another table. A 3rd table called a mapping table is required in order to implement such a relationship.

For example, one student can follow many courses and one course can have many students.

Student

AdNo	Name	DOB	sex
1110	Wimal Palitha	10/11/96	Male
2910	Sriya Menike	02/20/97	Female
2220	Lalitha Perera	12/03/96	Female
1203	B.Mohommed	05/12/97	Male
1230	T.Ramasinghe	21/11/97	Male

Courses

AdNo	CCode	Course Name	Duration
1110	220	Computer Science	3 Month
2910	220	Computer Science	3 Month
1110	106	Electronics NVQL3	4 Month
1230	106	Electronics NVQL3	4 Month
1230	220	Computer Science	3 Month

In the above student table Wimal Palitha , Sriya Menike and T. Ramasinghe follow Computer Science.

Palitha Wimal follows two courses (Computer Science, Electronics- NVQL3)

Creating a table relationship

Open the database → **Database Tools** tab → **Show/Hide** group → **Relationships** → If you have not yet defined any relationships, the **Show Table** dialog box automatically appears. If it does not appear, on the **Design** tab, in the **Relationships** group, click **Show Table** → Select the tables or queries → **Add** → **Close** the **Show Table** dialog box → Drag a field (typically the primary key) from one table to the common field (the foreign key) in the other table → **Edit Relationships** dialog box appears → To enforce referential integrity for this relationship, select the **Enforce Referential Integrity** check box → Click **OK** → Access draws a relationship line between the two tables. If you selected the **Enforce Referential Integrity** check box, the line appears thicker at each end. In addition, again only if you selected the **Enforce Referential Integrity** check box, the number **1** appears over the thick portion on one side of the relationship line, and the infinity symbol (∞) appears over the thick portion on the other side of the line, as shown in the figure

Modify the design of a table:

1. Modify the datasheet view:

□ Add a new field

Select the field which you want to modify → Right click on the field → **Insert column** → Rename the column as you want

□ Delete a field/record

Select the field which you want to modify → Right click on the field/record → **Delete field/Delete record**

□ Rename a field

Select the field which you want to modify → Right click on the field/record → **Rename field**

□ Add a new record

i. Select a row → Right click on the row header → **New Record** → You will be directed to the last row to add a new record

ii. Select a row → Click the **New (blank) record** button on the status bar

2. Modify the design view:

□ Add rows

Select the previous row/s where you want to add new row/s →

i. **Design** tab → **Tools** group → **Insert Rows**

ii. Right click on the selection → **Insert Rows**

□ Delete rows

i. Select the relevant row/s → Press **Delete** key → Click **Yes** to the message

ii. Select the relevant row/s → **Design** tab → **Tools** group → **Delete Rows** → Click **Yes** to the message

iii. Select the relevant row/s → Right click on the selection → **Delete Rows** → Click **Yes** to the message

Queries:

You can make **Queries** to view, change and analyze data in different ways. Also use them as the source of records for forms and reports.

Creating a Query:

Open a database which has tables or queries → **Create** tab → **Queries** group →

1. **Query Wizard** → **Simple Query Wizard** → **OK** → Select the **Table/s, Queries** → Select the fields you want for the query → **Next** → Select **Detail** or **Summary** → If you select **Summary**, apply **Summary Options** → **Next** → Give a **Title** to the query → Select **Open the query** or **Modify the query** → Finish

2. **Query Design** → **Add the Table/s, Query/s** → Close the **Show Table** dialog box → Add fields you want for the query → **Design** tab → **Results** group → **Run**

Save the Query:

1. **File** tab → **Save** → Type a name for the query → OK
2. **Click** the **Save** button on the **Quick Access toolbar** → Type a name for the query → OK
3. **Ctrl + S** → Type a name for the query → OK
4. Close the design view → Type a name for the query → OK

Searching Records in a Table or Query:

Open the datasheet view of the table or query → Place the cursor on the first cell of the relevant field or select the whole field → **Home** tab → **Find** group → **Find** → **Find and Replace** dialog box → **Find** tab → **Find what** → Type the word or number you want to find → Click **Find Next** until you finish searching

If you want to replace the word or number with another, click on **Replace** tab → **Replace with** → Type the word or number to be replaced with → Click **Replace / Replace All**

Searching for a Group of Records in a Query:

Open the design view of the query → Place the cursor in the **Criteria** row of the relevant field → Type the criteria → Open the datasheet view or run the query

Examples for Criterion-

- | | | |
|------------------------------------|---|--|
| ☐ >25and<75 | - | Records in between 25 and 75 |
| ☐ =35 | - | Records which are similar to 35 |
| ☐ Like S* | - | Records start with letter S |
| ☐ Is Null | - | Records in which the relevant field is empty |

Forms:

Creating a Form:

1. Open the table or query from which you want to create the form → **Create** tab → **Forms** group → **Form**
2. **Create** tab → **Forms** group → **Form Design** → **Design** tab → **Tools** group → Add **Existing Fields** → **Show all tables** → Open the relevant table → Drag and drop the fields you want for the form on to the form design view → Design the form as you want
3. **Create** tab → **Forms** group → **Blank Form** → Steps as earlier
4. **Create** tab → **Forms** group → **Form Wizard** → **Tables/Queries** → Select the table/s or query → Select the fields you want → **Next** → Select a **Layout** → **Next** → Type a title for the form → Select **Open the form** or **Modify the form** → **Finish**

5. Open the table or query from which you want to create the form → **Create** tab → **Forms** group → **More Forms** → **Multiple Items**
6. Open the table or query from which you want to create the form → **Create** tab → **Forms** group → **More Forms** → **Split Form**

Reports:

Report is the print preview of your table or query.

Creating a Report:

1. Open the table or query from which you want to create the report → **Create** tab → **Reports** group → **Report**
2. **Create** tab → **Reports** group → **Report Design** → **Design** tab → **Tools** group → Add **Existing Fields** → **Show all tables** → Open the relevant table → Drag and drop the fields you want for the report on to the report design view → Design the report as you want → **View** → **Report View**
3. **Create** tab → **Reports** group → **Blank Report** → Steps as earlier
4. **Create** tab → **Reports** group → **Report Wizard** → **Tables/Queries** → Select the table/s or query → Select the fields you want → **Next** → If you want, add a **Grouping Level** → **Next** → If you want, select a field to be sorted → **Next** → Select a **Layout** → **Next** → Type a title for the report → Select **Preview the report** or **Modify the report** → **Finish**